A Note on "FLRW Cosmology in Generic Gravity Theories": Extension to Non-Riemannian Geometries

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Abstract

In [1], the authors recently proved that, for the Friedmann-Lemaître-Robertson-Walker (FLRW) metric, the field equations of any generic metric gravity theory reduce to the perfect-fluid form. In this note, we demonstrate that this structural universality extends to some non-Riemannian geometries, specifically Einstein-Cartan theory [3] and Yano-Schrödinger cosmology [2]. We show that, provided the affine connection respects the spatial isotropy and homogeneity of the background, the resulting geometric corrections appearing in the field equations manifest strictly as effective perfect fluids.

1 Introduction

The study of generic gravity theories in the context of FLRW cosmology often relies on the restrictive symmetries of the metric. It has been established that for theories constructed from the metric and the Riemann tensor, the field equations are inevitably of the perfect fluid type $G_{\mu\nu} = T^{eff}_{\mu\nu}$ [1]. However, it is not yet known whether a similar result holds when one considers the presence of torsion and non-metricity, i.e., non-Riemannian geometries. We give two examples showing that the result of [1] holds even in non-Riemannian settings. The first example, from Yano-Schrödinger cosmology [2], has zero torsion and nonzero nonmetricity, whereas the second, from Einstein-Cartan theory [3], has nonzero torsion and zero nonmetricity.

2 The Yano-Schrödinger Case

In Yano-Schrödinger cosmology, the geometry is characterized by a length-preserving non-metricity vector π_μ. The generalized Einstein field equations in this framework take the form [2]:

$$G_{\mu\nu} + \mathcal{H}_{\mu\nu}(\pi) = \kappa T_{\mu\nu}, \tag{1}$$

where $G_{\mu\nu}$ is the standard Einstein tensor and $\mathcal{H}_{\mu\nu}(\pi)$ represents the geometric modification due to non-metricity:

$$\mathcal{H}_{\mu\nu}(\pi) = -\frac{5}{4}g_{\mu\nu}\nabla_\alpha\pi^\alpha + \frac{1}{2}\nabla_{(\mu}\pi_{\nu)} + \frac{5}{8}g_{\mu\nu}\pi^\alpha\pi_\alpha - \frac{1}{4}\pi_\nu\pi_\mu. \tag{2}$$

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To investigate the validity of [1], we impose the symmetries of the FLRW spacetime. Spatial isotropy and homogeneity require the non-metricity vector to be orthogonal to the spatial hypersurfaces. Thus, in comoving coordinates, it must take the form [2]:

$$\pi_\mu = (-\Pi(t), 0, 0, 0). \tag{3}$$

Substituting this symmetry-constrained vector into the geometric correction tensor $\mathcal{H}_{\mu\nu}$, we find that the off-diagonal terms vanish identically. The remaining diagonal components can be grouped strictly into an effective energy density ρ_{eff} and isotropic pressure p_{eff}.

Specifically, the time-time component yields an effective density contribution:

$$\rho_{eff} = \frac{1}{32\pi}\left(3\dot{\Pi} + 15H\Pi - \frac{3}{2}\Pi^2\right), \tag{4}$$

and the spatial components yield an effective pressure contribution:

$$p_{eff} = \frac{1}{32\pi}\left(-5\dot{\Pi} - 13H\Pi + \frac{5}{2}\Pi^2\right). \tag{5}$$

Consequently, the modified field equations can be rewritten entirely in the form of a standard General Relativistic system coupled to an effective perfect fluid:

$$G_{\mu\nu} = \kappa\left[(\rho_m + \rho_{eff})u_\mu u_\nu + (p_m + p_{eff})h_{\mu\nu}\right]. \tag{6}$$

This demonstrates that [1] holds: the non-metric geometric terms are algebraically indistinguishable from a perfect fluid source, provided the non-metricity respects the background symmetries.

3 The Einstein-Cartan Case

In the Einstein-Cartan framework, the geometry remains metric-compatible, but the connection possesses a non-vanishing torsion tensor $T^\lambda_{\mu\nu}$ sourced by the intrinsic spin of matter. The generalized field equations in this theory are given by [3]:

$$G_{\mu\nu} - \Lambda g_{\mu\nu} = \kappa\left(T_{\mu\nu} + \theta_{\mu\nu}\right), \tag{7}$$

where $T_{\mu\nu}$ is the canonical stress-energy tensor and $\theta_{\mu\nu}$ represents the geometric correction terms arising from the spin-torsion interaction.

To verify [1], we model the matter source as a Weyssenhoff fluid, which generalizes the perfect fluid to include non-vanishing spin density $S_{\mu\nu}$. Consistent with the cosmological principle of the FLRW background, we assume the macroscopic average of the spin orientation vanishes ($\langle S_{\mu\nu}\rangle = 0$), but the spin-squared contribution $\langle S^2\rangle$ remains non-zero and physically significant.

Under these symmetry conditions, the correction tensor $\theta_{\mu\nu}$ does not introduce off-diagonal shear or anisotropic stress. Instead, it modifies the diagonal components of the field equations. Specifically, the expanded field equations take the form [3]:

$$G_{\mu\nu} = \kappa\left(\rho + p - \frac{\kappa}{2}S^2\right)u_\mu u_\nu + \kappa\left(p - \frac{\kappa}{4}S^2\right)g_{\mu\nu}, \tag{8}$$

where we have taken the isotropic limit $p_r = p_t = p$.

By inspecting the algebraic structure of this equation, we can identify the geometric corrections as effective shifts to the thermodynamic variables:

$$p_{eff} = p - \frac{\kappa}{4}S^2, \tag{9}$$

$$\rho_{eff} = \rho - \frac{\kappa}{4}S^2. \tag{10}$$

Substituting these effective quantities back into the field equations yields the standard perfect fluid form:

$$G_{\mu\nu} = \kappa \left[(\rho_{eff} + p_{eff}) u_\mu u_\nu + p_{eff} g_{\mu\nu} \right]. \tag{11}$$

This again confirms that [1] is robust in the presence of torsion: the geometric modifications induced by spin-torsion coupling respect the background FLRW symmetries and manifest strictly as an effective perfect fluid with modified equation of state.

4 Conclusion

In this note, we have investigated the stability of the perfect fluid form in FLRW cosmologies extended with non-Riemannian defects. While the original result [1] established that field equations derived from generic Riemannian invariants reduce to the perfect fluid form, the landscape of metric-affine gravity is significantly broader. Rather than attempting a generic proof for all possible non-Riemannian invariants, we focused on two physically significant extensions: Yano-Schrödinger gravity (pure non-metricity) and Einstein-Cartan theory (pure torsion).

We have demonstrated that in both cases, the imposition of spatial isotropy forces the geometric correction tensors to diagonalize. Possible lines of future inquiry could include formally extending this analysis to the general case, where torsion and non-metricity are present simultaneously, to verify if the perfect fluid form is preserved under coupled defects. Additionally, anisotropic cosmological models (such as Bianchi classifications) could be investigated to determine whether the coupling between shear and torsion introduces fundamentally new effective stress-energy components that defy the perfect fluid description.

References

[1] M. Gürses and Y. Heydarzade, "FLRW cosmology in generic gravity theories," *Eur. Phys. J. C* **80**, 1061 (2020).

[2] L. Csillag, A. Agashe, and D. Iosifidis, "Schrödinger connections: from mathematical foundations towards Yano–Schrödinger cosmology," *Class. Quantum Grav.* **41**, 235005 (2024).

[3] M. R. Mehdizadeh and A. H. Ziaie, "Einstein-Cartan wormhole solutions," *Phys. Rev. D* **95**, 064049 (2017).